

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

Course Code: ITA0609

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CO4 ASSESSMENT TOOL 2 – INDUDTRY PROBLEM SOLVING

1. A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using K-Nearest Neighbour or Case Based Learning. Explain how similarity between users is measured, how recommendations are generated, and how the system adapts to changing user behavior over time.

Title

Data Plan Recommendation Using K-Nearest Neighbour (KNN)

Aim

To design a KNN-based recommendation system that recommends suitable telecom data plans based on customer usage patterns.

Problem Identification

- Customers have different data and calling requirements.
- A single plan may not be suitable for all customers.
- Customer usage patterns can be used to find similar users.
- KNN can recommend plans based on the behaviour of similar customers.
- The system can adapt when customer usage changes.

Proposed Solution

- Collect customer data such as data usage, call duration, SMS usage, and browsing hours.
- Normalize the input features.
- Calculate the distance between the target customer and existing customers.
- Select the K nearest customers.
- Recommend the plan most commonly used by these customers.

Similarity Measurement

Euclidean distance is used : Smaller distance = higher similarity

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}$$

Example

Suppose K = 3:

Customer	Data Usage	Call Minutes	Plan
A	8 GB	200	10 GB
B	9 GB	180	10 GB
C	15 GB	250	20 GB
D	7 GB	190	10 GB
Target	9 GB	195	?

The three nearest customers are A, B and D.

- A → 10 GB
- B → 10 GB
- D → 10 GB

Recommendation = 10 GB Data Plan

Adaptation to Changing Behaviour

- Continuously collect recent usage data.
- Update the customer database regularly.
- Give importance to recent usage patterns.
- Recalculate nearest neighbours when usage changes.
- Recommend a new plan when the customer's requirements increase or decrease.

Result

- Personalized data plans are recommended.
- Customer satisfaction can be improved.
- Unnecessary data expenses can be reduced.
- Recommendations are based on actual usage.
- The system adapts to changing customer behaviour.

Conclusion

KNN provides a simple and effective personalized recommendation system by identifying customers with similar usage patterns and recommending the plan preferred by their nearest neighbours.

2. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.

Title

Electricity Consumption Prediction Using RBF Network

Aim

To develop an **RBF Network-based prediction system** for estimating electricity consumption using time, weather, and appliance usage data.

Problem Identification

- Electricity consumption changes throughout the day.
- Weather conditions affect energy demand.
- Appliance usage creates nonlinear consumption patterns.
- Measurement data may contain noise.
- Seasonal variations make accurate prediction difficult.

Proposed Solution

- Collect historical electricity consumption data.
- Use **time, temperature, humidity, and appliance usage** as inputs.
- Train an RBF Network using historical data.
- RBF hidden units learn local consumption patterns.
- Predict future electricity consumption.

Input Parameters

- Time of day
- Temperature
- Humidity
- Day of week

- Appliance usage
- Previous electricity consumption

RBF Network

Input Layer → RBF Hidden Layer → Output Layer

The RBF function is:

$$\phi(x) = e^{-\frac{\|x-c\|^2}{2\sigma^2}}$$

where **c** represents the centre of a local pattern and **σ** controls its spread.

Example

Suppose the system receives:

Parameter	Value
Time	7:00 PM
Temperature	32°C
Humidity	65%
AC Usage	4 hours
Previous Consumption	8 kWh

The trained RBF network processes these inputs.

Predicted electricity consumption = 10.5 kWh

This prediction can help the company prepare for increased evening demand.

Local Learning

- Each RBF neuron focuses on a particular input region.
- Different neurons learn different consumption conditions.
- One neuron may learn **summer afternoon** patterns.
- Another may learn **winter nighttime** patterns.
- This improves prediction for local conditions.

Handling Noisy Data

- Remove abnormal measurements.
- Normalize input data.

- Use suitable RBF spread values.
- Apply regularization to reduce overfitting.
- Train using sufficient historical data.

Handling Seasonal Data

- Include month and weather information.
- Train using data from different seasons.
- Learn separate patterns for summer, winter, and monsoon periods.
- Periodically update the model with new consumption data.

Result

- Electricity consumption can be predicted accurately.
- Local and nonlinear patterns are captured.
- Noise has less influence on predictions.
- Seasonal variations can be handled.
- Helps improve energy planning and distribution.

Conclusion

An RBF Network provides an effective **local-learning approach** for electricity forecasting because it can model nonlinear, noisy, and seasonal consumption patterns.